

The Cascade Curvature Operator under Refinement: Schur–Complement Identity and Energy Intertwining

A predecessor to the cascade-closure mechanism paper

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Abstract

We discharge two of the four consumer-API obligations in the cascade-closure mechanism paper [1] for the finite-level P3/P4 refinement tower of [3, 2]. With the scaled curvature operator $\tilde{A}_n := 2^n A_n^{\text{vertex}}$, where $A_n^{\text{vertex}} = d_n^* d_n$ is the source’s vertex graph-Laplacian and 2^n matches the source’s factor- $\frac{1}{2}$ coboundary refinement relation $p^1 d_{n+1} = \frac{1}{2} d_n p^0$, the Schur complement of $\tilde{A}_{n+1}$ with respect to the (boundary, interior) decomposition induced by edge-midpoint subdivision equals $\tilde{A}_n$ exactly: $\text{Schur}(\tilde{A}_{n+1}) = \tilde{A}_n$. This identity is verified analytically and numerically on two concrete refinement towers (single edge to one-midpoint subdivision; triangle to barycentric edge-subdivision) and yields, on the boundary-energy sector, the cascade-mechanism residual-monotonicity hypothesis $\tilde{R}_n(p^0 \psi) \leq \tilde{R}_{n+1}(\psi)$ for every $\psi \in X_{n+1}^0$, with equality if and only if ψ is the harmonic extension of $p^0 \psi$. Strict operator-level refinement-compatibility $p^0 \tilde{A}_{n+1} = \tilde{A}_n p^0$ remains false; the defect operator vanishes precisely on the harmonic-extension subspace and is computed explicitly for the worked towers. The result discharges cascade-mechanism’s obligation (O3) (residual monotonicity) for the P3/P4 tower and reformulates obligation (O2) (flow intertwining) at the quadratic-form / Schur level, where it holds, rather than at the strict operator level, where it does not.

Scope

- This is a narrow technical paper. It contains no new mechanism, no programme framing, no consciousness or cosmology claim. It is the source-side companion to [1].
- Its job is to discharge as many of the consumer-API obligations (O0)–(O3) of [1] §3.4 as the imported P3/P4 finite-level tower will support.
- Two of the four obligations are closed here, in the precise form recorded in §7; one is closed only at the boundary-energy / Schur level rather than at the operator/flow level; one is recorded as essentially trivial (kernel of a connected graph Laplacian).

1 Introduction and source-state recap

The cascade-closure mechanism paper [1] defines cascade closure as a rung-indexed closure-residual / projection-stack event, proves a conditional residual-zero / fixed-point propagation proposition, and supplies a consumer API listing four obligations that successor papers must discharge to invoke cascade closure as a load-bearing premise:

- (O0) Standing analytic / admissibility hypotheses.
- (O1) Top-rung fixed point and convergence.
- (O2) Flow intertwining $\pi_{k+1,k}^\# \circ \Psi_{k+1}^t = \Psi_k^t \circ \pi_{k+1,k}^\#$.
- (O3) Residual monotonicity $R_k(\pi_{k+1,k}^\# \psi) \leq R_{k+1}(\psi)$.

[1] §3.5 records the per-obligation status in the imported P3/P4 finite-level tower of [3, 2]: (O0) is satisfied at the finite-dimensional level; (O1)–(O3) are open in the source state, with (O2) closed only for the source’s mass-only operator $L_n = -\gamma C_n$ at $\alpha = \beta = 0$, not for the curvature/residual operator A_n that the cascade-mechanism flow naturally uses. [1] §3.5 names the minimal upstream upgrade required and observes that the role of [1] is to make the obligations visible as the programme’s actual dependency gate, not to discharge them.

The present paper is that source-side discharge, applied to the imported P3/P4 finite-level tower with the curvature operator A_n rather than the source’s mass-only L_n . We do not redefine the cascade-closure mechanism, do not modify [1]’s consumer API, and do not introduce any programme-level claim. We supply a single source-side identity and exhibit its consequences for (O2) and (O3).

1.1 Source objects (recap, no new content)

We use the notation of [2] verbatim. At each refinement level n the source provides a graph $G_n^\bullet$ with vertex set $V(G_n^\bullet)$ and edge set $E(G_n^\bullet)$, and the chain spaces $X_n^0 := \mathbb{R}^{V(G_n^\bullet)}$, $X_n^1 := \mathbb{R}^{E(G_n^\bullet)}$, $X_n := X_n^0 \oplus X_n^1$, together with the discrete coboundary $d_n : X_n^0 \rightarrow X_n^1$ and its adjoint $d_n^* : X_n^1 \rightarrow X_n^0$ in the standard ℓ^2 inner products. The vertex-block curvature operator is

$$A_n^{\text{vertex}} := d_n^* d_n \in \mathbb{R}^{V(G_n^\bullet) \times V(G_n^\bullet)}, \quad (1)$$

the standard graph-Laplacian on $V(G_n^\bullet)$.

The bonding maps are the pair $p_{n+1,n} = (p_{n+1,n}^0, p_{n+1,n}^1)$, where $p_{n+1,n}^0 : X_{n+1}^0 \rightarrow X_n^0$ is vertex-restriction along the canonical inclusion $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$ and $p_{n+1,n}^1 : X_{n+1}^1 \rightarrow X_n^1$ is the parent-fibre average over the canonical edge-parent map $\pi_{n+1,n}^E : E(G_{n+1}^\bullet) \rightarrow E(G_n^\bullet)$ (each level- n edge subdivides into exactly two level- $(n+1)$ edges; the average is by factor $\frac{1}{2}$ on each child).

The source’s load-bearing identity is the coboundary refinement relation [2] Proposition **coboundary-refinement** (lines 896–937):

$$p_{n+1,n}^1 d_{n+1} = \frac{1}{2} d_n p_{n+1,n}^0. \quad (2)$$

We use (2) as a black box.

1.2 What this paper proves

The paper’s central object is the scaled vertex curvature operator:

Definition 1.1 (Scaled curvature operator). $\tilde{A}_n := 2^n A_n^{\text{vertex}}$, defined on X_n^0 . The corresponding scaled residual is $\tilde{R}_n(\phi) := \frac{1}{2} \langle \phi, \tilde{A}_n \phi \rangle = 2^{n-1} \langle \phi, A_n^{\text{vertex}} \phi \rangle$.

The scaling factor 2^n is the unique factor that compensates the refinement-induced halving in (2) once one passes from the coboundary level to the curvature (d^*d) level. The main results are an exact Schur-complement identity for $\tilde{A}_n$ under boundary–interior decomposition (Theorem 2.1), a clean energy-intertwining inequality on the harmonic-extension sector that discharges cascade-mechanism’s (O3) (Theorem 3.1), and an explicit defect identity showing that strict operator-level refinement-compatibility for $\tilde{A}_n$ remains false (Proposition 4.1).

2 Schur-complement identity

Fix a level n and a finite refinement $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$ such that $|V(G_{n+1}^\bullet)| - |V(G_n^\bullet)|$ new “mid-point” vertices are added (each created as the midpoint of one existing edge of $G_n^\bullet$ that was subdivided to produce $G_{n+1}^\bullet$). Partition the vertex set $V(G_{n+1}^\bullet) = B \sqcup I$ into

- $B := V(G_n^\bullet)$, the *boundary* vertices (those retained from level n);
- $I := V(G_{n+1}^\bullet) \setminus V(G_n^\bullet)$, the *interior* or *new* vertices.

Block-decompose A_{n+1}^{vertex} in this (B, I) ordering:

$$A_{n+1}^{\text{vertex}} = \begin{pmatrix} A_{BB} & A_{BI} \\ A_{IB} & A_{II} \end{pmatrix}, \quad (3)$$

where $A_{BB} \in \mathbb{R}^{B \times B}$, $A_{BI} = A_{IB}^T \in \mathbb{R}^{B \times I}$, $A_{II} \in \mathbb{R}^{I \times I}$.

By construction A_{n+1}^{vertex} is the graph-Laplacian on $V(G_{n+1}^\bullet)$. The harmonic extension $H_{n+1} : X_n^0 \rightarrow X_{n+1}^0$ is the unique linear map sending $f \in X_n^0$ to the level- $(n+1)$ state with prescribed boundary values f and interior values $f_I = -A_{II}^{-1}A_{IB}f$, equivalently the unique minimiser of the level- $(n+1)$ energy $\frac{1}{2}\langle \cdot, A_{n+1}^{\text{vertex}} \cdot \rangle$ subject to the boundary constraint $p_{n+1,n}^0 \cdot = f$. We assume A_{II} is invertible (the standard finite-dimensional case in [3, 2]).

The Schur complement of A_{n+1}^{vertex} with respect to the partition (B, I) is

$$\text{Schur}(A_{n+1}^{\text{vertex}}) := A_{BB} - A_{BI}A_{II}^{-1}A_{IB} \in \mathbb{R}^{B \times B}. \quad (4)$$

Theorem 2.1 (Schur-complement halving). *For the cascade refinement tower with $A_n^{\text{vertex}} = d_n^* d_n$, where the refinement is the edge-midpoint subdivision scheme of [2] (each level- n edge $e = (u, v)$ is subdivided by inserting one new midpoint vertex m_e of degree 2 in $G_{n+1}^\bullet$ with the two child edges (u, m_e) and (m_e, v) , and no other level- $(n+1)$ edges incident to midpoints), the Schur complement of A_{n+1}^{vertex} under the (boundary, interior) partition (3) is exactly half the level- n Laplacian:*

$$\text{Schur}(A_{n+1}^{\text{vertex}}) = \frac{1}{2} A_n^{\text{vertex}}. \quad (5)$$

Equivalently, with $\tilde{A}_n := 2^n A_n^{\text{vertex}}$:

$$\text{Schur}(\tilde{A}_{n+1}) = \tilde{A}_n. \quad (6)$$

Proof. We proceed in three steps: identify the Schur complement with a harmonic energy, decompose that energy via the source’s coboundary refinement relation, and verify directly that the harmonic extension saturates the resulting lower bound.

Step 1. For any positive-semi-definite A_{n+1}^{vertex} with invertible interior block A_{II} , the Schur complement and the harmonic-extension energy agree: $\langle f, \text{Schur}(A_{n+1}^{\text{vertex}})f \rangle = \min_{\psi: p^0 \psi = f} \langle \psi, A_{n+1}^{\text{vertex}} \psi \rangle = \langle H_{n+1}f, A_{n+1}^{\text{vertex}} H_{n+1}f \rangle$ for every $f \in X_n^0$. This is standard: for $\psi = (f, f_I)$ in block form, $\langle \psi, A_{n+1}^{\text{vertex}} \psi \rangle = \langle f, A_{BB}f \rangle + 2\langle f_I, A_{IB}f \rangle + \langle f_I, A_{II}f_I \rangle$ is minimised in f_I at $f_I^* = -A_{II}^{-1}A_{IB}f$ (since A_{II} is positive definite), with minimum value $\langle f, (A_{BB} - A_{BI}A_{II}^{-1}A_{IB})f \rangle$, and $H_{n+1}f := (f, f_I^*)$ by definition of harmonic extension.

Step 2. Let $f \in X_n^0$ and let $\psi \in X_{n+1}^0$ satisfy $p^0 \psi = f$. Apply the coboundary refinement relation (2): $p^1 d_{n+1} \psi = \frac{1}{2} d_n p^0 \psi = \frac{1}{2} d_n f$, which is fixed independently of ψ (subject to $p^0 \psi = f$). The averaging-by- $\frac{1}{2}$ structure of p^1 on a 2-element fibre yields the calibration identity

$$p^1(p^1)^* = \frac{1}{2} I_{X_n^1}, \quad (7)$$

which is verified directly: for any $\omega \in X_n^1$ and any level- n edge e with two children e_1, e_2 , one has $((p^1)^* \omega)(e_i) = \frac{1}{2} \omega(e)$ (the unique adjoint under standard ℓ^2 inner products) and $(p^1(p^1)^* \omega)(e) =$

$\frac{1}{2} \cdot \frac{1}{2} \omega(e) + \frac{1}{2} \cdot \frac{1}{2} \omega(e) = \frac{1}{2} \omega(e)$. Hence the orthogonal projection of any $\xi \in X_{n+1}^1$ onto $\text{range}((p^1)^*)$ is $P\xi = (p^1)^*((p^1(p^1)^*)^{-1}p^1\xi) = 2(p^1)^*p^1\xi$, and X_{n+1}^1 decomposes orthogonally as $\text{range}((p^1)^*) \oplus \ker p^1$. Apply this decomposition to $\xi = d_{n+1}\psi$:

$$d_{n+1}\psi = (p^1)^*d_nf + r_\psi, \quad r_\psi \in \ker p^1, \quad (8)$$

where the first term comes from $P(d_{n+1}\psi) = 2(p^1)^*(p^1d_{n+1}\psi) = 2(p^1)^*(\frac{1}{2}d_nf) = (p^1)^*d_nf$. Compute the norm using (7):

$$\|(p^1)^*d_nf\|^2 = \langle (p^1)^*d_nf, (p^1)^*d_nf \rangle = \langle d_nf, p^1(p^1)^*d_nf \rangle = \frac{1}{2}\|d_nf\|^2 = \frac{1}{2}\langle f, A_n^{\text{vertex}}f \rangle.$$

By orthogonality of the decomposition (8),

$$\langle \psi, A_{n+1}^{\text{vertex}}\psi \rangle = \|d_{n+1}\psi\|^2 = \frac{1}{2}\langle f, A_n^{\text{vertex}}f \rangle + \|r_\psi\|^2 \geq \frac{1}{2}\langle f, A_n^{\text{vertex}}f \rangle. \quad (9)$$

Step 3. We exhibit one specific ψ with $r_\psi = 0$, hence equality in (9): the harmonic extension itself. By the edge-midpoint hypothesis of the theorem statement, every interior vertex $m_e \in I$ has degree exactly 2 in $G_{n+1}^\bullet$, its only neighbours being the two endpoints u, v of its parent edge $e \in E(G_n)$; hence the harmonic-extension formula $f_I = -A_{II}^{-1}A_{IB}f$ at m_e reduces to the elementary two-neighbour average $(H_{n+1}f)(m_e) = \frac{1}{2}(f(u) + f(v))$. Compute $d_{n+1}(H_{n+1}f)$ at the two child edges of e :

$$\begin{aligned} (d_{n+1}(H_{n+1}f))(u, m_e) &= (H_{n+1}f)(m_e) - (H_{n+1}f)(u) = \frac{1}{2}(f(u) + f(v)) - f(u) = \frac{1}{2}(f(v) - f(u)) = \frac{1}{2}(d_nf)(e), \\ (d_{n+1}(H_{n+1}f))(m_e, v) &= (H_{n+1}f)(v) - (H_{n+1}f)(m_e) = f(v) - \frac{1}{2}(f(u) + f(v)) = \frac{1}{2}(f(v) - f(u)) = \frac{1}{2}(d_nf)(e). \end{aligned}$$

On the other hand, $((p^1)^*d_nf)(u, m_e) = \frac{1}{2}(d_nf)(e) = ((p^1)^*d_nf)(m_e, v)$ by definition of $(p^1)^*$ on a two-element fibre. Hence $d_{n+1}(H_{n+1}f) = (p^1)^*d_nf$ on every level- $(n+1)$ edge, i.e., $r_{H_{n+1}f} = 0$.

Combining Steps 1–3:

$$\langle f, \text{Schur}(A_{n+1}^{\text{vertex}})f \rangle = \langle H_{n+1}f, A_{n+1}^{\text{vertex}}H_{n+1}f \rangle = \frac{1}{2}\langle f, A_n^{\text{vertex}}f \rangle + \|r_{H_{n+1}f}\|^2 = \frac{1}{2}\langle f, A_n^{\text{vertex}}f \rangle.$$

This holds for every $f \in X_n^0$. Since $\text{Schur}(A_{n+1}^{\text{vertex}})$ and A_n^{vertex} are both symmetric, polarisation of the quadratic form gives the operator identity (5). The scaled form (6) follows by linearity: Schur is a quadratic operation but its dependence on the overall scale of the input matrix is linear in that scale, so $\text{Schur}(\tilde{A}_{n+1}) = \text{Schur}(2^{n+1}A_{n+1}^{\text{vertex}}) = 2^{n+1}\text{Schur}(A_{n+1}^{\text{vertex}}) = 2^{n+1} \cdot \frac{1}{2}A_n^{\text{vertex}} = 2^n A_n^{\text{vertex}} = \tilde{A}_n$. $\square$

Remark 2.2 (Numerical verification). Theorem 2.1 is verified to exact rational precision in §6 for two finite refinement instances: the single edge with one-midpoint subdivision (Tower I, $|V(G_0^\bullet)| = 2$, $|V(G_1^\bullet)| = 3$) and the triangle with three-midpoint barycentric subdivision (Tower II, $|V(G_0^\bullet)| = 3$, $|V(G_1^\bullet)| = 6$). The numerical witness is `papers/cascade-refinement-compat/scripts/explore_defect.py`; all linear algebra is exact `Fraction` arithmetic, with no floating-point error.

3 Energy intertwining and discharge of (O3)

We now apply Theorem 2.1 to the cascade-mechanism consumer-API obligations. The cascade-mechanism residual functional in this paper's setting is $\tilde{R}_n(\phi) := \frac{1}{2}\langle \phi, \tilde{A}_n\phi \rangle$.

Theorem 3.1 (Energy intertwining; discharge of (O3)). *For every $\psi \in X_{n+1}^0$,*

$$\tilde{R}_n(p_{n+1,n}^0\psi) \leq \tilde{R}_{n+1}(\psi), \quad (10)$$

with equality if and only if $\psi = H_{n+1}(p_{n+1,n}^0\psi)$ (i.e. ψ is the harmonic extension of its boundary).

Proof. Decompose $\psi = H_{n+1}(p_{n+1,n}^0\psi) + \psi^\perp$, where $\psi^\perp := \psi - H_{n+1}(p_{n+1,n}^0\psi)$ vanishes on $B = V(G_n^\bullet)$ (since $p_{n+1,n}^0 H_{n+1}(p_{n+1,n}^0\psi) = p_{n+1,n}^0\psi$). The energy quadratic form decomposes as

$$\begin{aligned} \langle \psi, A_{n+1}^{\text{vertex}} \psi \rangle &= \langle H_{n+1}(p^0\psi), A_{n+1}^{\text{vertex}} H_{n+1}(p^0\psi) \rangle \\ &\quad + 2\langle H_{n+1}(p^0\psi), A_{n+1}^{\text{vertex}} \psi^\perp \rangle + \langle \psi^\perp, A_{n+1}^{\text{vertex}} \psi^\perp \rangle. \end{aligned}$$

The cross term vanishes: writing A_{n+1}^{vertex} in block form (3) and noting that $\psi^\perp$ has zero B -component, one has $A_{n+1}^{\text{vertex}} \psi^\perp = (A_{BI} \psi_I^\perp, A_{II} \psi_I^\perp)$, and $H_{n+1}(p^0\psi) = (p^0\psi, -A_{II}^{-1} A_{IB} p^0\psi)$; the cross-pairing computes to $\langle p^0\psi, A_{BI} \psi_I^\perp \rangle + \langle -A_{II}^{-1} A_{IB} p^0\psi, A_{II} \psi_I^\perp \rangle = \langle p^0\psi, A_{BI} \psi_I^\perp \rangle - \langle p^0\psi, A_{BI} \psi_I^\perp \rangle = 0$. The first surviving term equals $\langle p^0\psi, \text{Schur}(A_{n+1}^{\text{vertex}}) p^0\psi \rangle = \frac{1}{2} \langle p^0\psi, A_n^{\text{vertex}} p^0\psi \rangle$ by Theorem 2.1. The third surviving term is $\langle \psi_I^\perp, A_{II} \psi_I^\perp \rangle \geq 0$ since A_{II} is positive semi-definite (a principal submatrix of the positive-semi-definite A_{n+1}^{vertex}). Hence

$$\langle \psi, A_{n+1}^{\text{vertex}} \psi \rangle \geq \frac{1}{2} \langle p^0\psi, A_n^{\text{vertex}} p^0\psi \rangle.$$

Multiplying both sides by 2^n : $2^n \langle \psi, A_{n+1}^{\text{vertex}} \psi \rangle \geq 2^{n-1} \langle p^0\psi, A_n^{\text{vertex}} p^0\psi \rangle$, which after dividing by 2 and re-grouping factors becomes $\langle \psi, \tilde{A}_{n+1} \psi \rangle / 2 \geq \langle p^0\psi, \tilde{A}_n p^0\psi \rangle / 2$, i.e. $\tilde{R}_{n+1}(\psi) \geq \tilde{R}_n(p^0\psi)$, which is (10). Equality requires $\psi^\perp = 0$ (since A_{II} is positive definite by invertibility), i.e. ψ is the harmonic extension of its boundary. $\square$

Corollary 3.2 ((O3) is closed for the P3/P4 tower with $\tilde{A}_n$). *The cascade-mechanism residual-monotonicity hypothesis [1] (O3) holds for the P3/P4 finite-level refinement tower with the scaled curvature residual $\tilde{R}_n(\phi) = \frac{1}{2} \langle \phi, \tilde{A}_n \phi \rangle$, where $\tilde{A}_n = 2^n A_n^{\text{vertex}}$ and $A_n^{\text{vertex}} = d_n^* d_n$ is the source's vertex graph-Laplacian.*

4 Strict operator-level refinement-compatibility fails

Theorem 2.1 closes the energy-level refinement question. The strict operator-level identity $p_{n+1,n}^0 \tilde{A}_{n+1} = \tilde{A}_n p_{n+1,n}^0$ remains false, with explicit defect.

Proposition 4.1 (Strict operator intertwining fails; defect vanishes on harmonic extensions). *The strict identity $p_{n+1,n}^0 \tilde{A}_{n+1} = \tilde{A}_n p_{n+1,n}^0$ does not hold in general, and equivalently $p_{n+1,n}^0 A_{n+1}^{\text{vertex}} \neq \frac{1}{2} A_n^{\text{vertex}} p_{n+1,n}^0$ as operators $X_{n+1}^0 \rightarrow X_n^0$. The defect operator*

$$D_n := p_{n+1,n}^0 A_{n+1}^{\text{vertex}} - \frac{1}{2} A_n^{\text{vertex}} p_{n+1,n}^0 \quad (11)$$

satisfies the following:

- (i) $D_n \psi = 0$ whenever $\psi \in \text{range}(H_{n+1})$, i.e. whenever ψ is the harmonic extension of its own boundary $p^0\psi$;
- (ii) D_n is generically non-zero on X_{n+1}^0 outside $\text{range}(H_{n+1})$. In particular, D_n does not vanish on the subspace $\{\psi : \psi_I = 0\}$.

Proof. (i) Apply D_n to a harmonic extension $\psi = H_{n+1}f$, $f := p^0\psi \in X_n^0$. Then $p^0\psi = f$, and the energy-form identity inside the proof of Theorem 2.1 yields $\langle \eta, p^0 A_{n+1}^{\text{vertex}} H_{n+1}f \rangle = \langle \eta, \frac{1}{2} A_n^{\text{vertex}} f \rangle$ for every $\eta \in X_n^0$. (This is the inner product $\langle \eta, \cdot \rangle$ pulled back through the Schur identity: the right side is $\frac{1}{2} \langle \eta, A_n^{\text{vertex}} f \rangle$, the left side is $\langle p_0^* \eta, A_{n+1}^{\text{vertex}} H_{n+1}f \rangle$ where p_0^* is the obvious zero-extension to interior; both equal $\frac{1}{2} \langle f, A_n^{\text{vertex}} \eta \rangle$ by symmetry of A_n^{vertex} and the Step 3 calculation in the proof of Theorem 2.1.) Hence $p^0 A_{n+1}^{\text{vertex}} H_{n+1}f = \frac{1}{2} A_n^{\text{vertex}} p^0 H_{n+1}f$, which is exactly $D_n \cdot H_{n+1}f = 0$.

(ii) On any state $\phi \in X_{n+1}^0$ with $\phi_I = 0$, i.e. $\phi = (\phi_B, 0)$ in block form, one computes $p^0 A_{n+1}^{\text{vertex}} \phi = A_{BB} \phi_B$ (the (B, I) block on a state with $\phi_I = 0$). On the other side, $\frac{1}{2} A_n^{\text{vertex}} p^0 \phi = \frac{1}{2} A_n^{\text{vertex}} \phi_B$. The block A_{BB} is the level- $(n+1)$ Laplacian restricted to boundary vertices, which records edges from B to neighbours both in B and in I ; for the edge-midpoint refinement scheme the endpoints of every level- n edge $e \in E(G_n^\bullet)$ acquire a new neighbour at level $n+1$ (the midpoint m_e), so A_{BB} differs from $\frac{1}{2} A_n^{\text{vertex}}$ in general. The explicit defect $D_n \phi$ is computed for two concrete towers in §6; in both, $D_n|_{\{\phi_I=0\}}$ is non-zero. $\square$

Remark 4.2 (Geometric interpretation). The conclusion (i) of Proposition 4.1 is the operator-level shadow of Theorem 2.1: harmonic extensions are exactly the level- $(n+1)$ states whose interior values do not contribute to the boundary Laplacian action under p^0 , hence the only states for which the strict scaled identity holds. On non-harmonic states, the defect measures the “midpoint deficit” $\psi_I - H_{n+1}(p^0 \psi)|_I$, i.e. how far the interior values of ψ are from being harmonic averages of the boundary values.

Remark 4.3 (Energy vs. flow level). The Schur identity Theorem 2.1 and its energy consequence Theorem 3.1 hold at the *quadratic-form / boundary-energy* level. Strict *operator-level* intertwining $p^0 \tilde{A}_{n+1} = \tilde{A}_n p^0$ fails by Proposition 4.1, and consequently the strict *flow-level* intertwining $p^0 \circ e^{-t\tilde{A}_{n+1}} = e^{-t\tilde{A}_n} \circ p^0$ also fails. Cascade-mechanism’s obligation (O2), as stated, is at the flow level. The natural source-supported reformulation of (O2) is the Schur-energy version, which Theorem 2.1 discharges. In the cascade-mechanism reformulation table of §7 we record this as a *partial* discharge of (O2).

5 Kernel of A_n^{vertex} and (O1)

For completeness we record the status of cascade-mechanism’s obligation (O1) (top-rung residual-zero fixed point and convergence) in the imported P3/P4 finite-level tower.

Proposition 5.1 (Kernel and convergence of $\tilde{A}_N$). *For any finite level N , $\tilde{A}_N$ is positive semi-definite on X_N^0 and its kernel $\ker \tilde{A}_N = \ker A_N^{\text{vertex}}$ consists exactly of the locally-constant functions on the connected components of $G_N^\bullet$. If $G_N^\bullet$ is connected, $\dim \ker \tilde{A}_N = 1$ (constants only).*

For every initial datum $\phi^{(0)} \in X_N^0$, the gradient flow $\Psi_N^t := e^{-t\tilde{A}_N}$ converges as $t \rightarrow \infty$ to the orthogonal projection of $\phi^{(0)}$ onto $\ker \tilde{A}_N$. This is automatic in finite dimensions by spectral decomposition.

In particular, the trivial state $\phi_N^ = 0$ is always a residual-zero fixed point, and the cascade-mechanism obligation (O1) “nontrivial-selected-state existence” reduces to whether $\ker \tilde{A}_N \neq \{0\}$ (yes, in the connected case the kernel is one-dimensional) and whether the projection of the initial datum onto the kernel is non-zero (which is generic but not automatic).*

Proof. $A_N^{\text{vertex}} = d_N^* d_N$ is positive semi-definite by construction, with kernel $\ker d_N$, the locally-constant functions on the connected components of $G_N^\bullet$. The convergence statement for $e^{-tA_N^{\text{vertex}}}$ is the standard finite-dimensional functional-calculus result for a self-adjoint positive-semi-definite operator. Multiplying by the positive scalar 2^N does not change the kernel or the convergence behaviour. $\square$

Remark 5.2 (Selection rule is not part of cascade closure). [1] §5 explicitly states that selection of the top-rung fixed point is not provided by cascade closure and is the successor paper’s own task. The kernel description above records what cascade closure inherits at the top rung; the selection rule by which a particular non-zero element of $\ker \tilde{A}_N$ is preferred over the zero state is a question for downstream papers, and outside the scope of the present source-side discharge.

6 Worked finite refinement towers

This section gives explicit matrix-level computations on two concrete finite refinement towers, verifying Theorems 2.1 and 3.1 and Proposition 4.1 to exact rational precision. The numerical witness is the script `papers/cascade-refinement-compat/scripts/explore_defect.py`; its output is preserved in `papers/cascade-refinement-compat/repro/expected_outputs/`.

6.1 Tower I: single edge $\rightarrow$ one-midpoint subdivision

Let $G_0^\bullet$ be the graph on $V(G_0^\bullet) = \{u, v\}$ with the single edge $e = (u, v)$. Refine by inserting the midpoint m of e , splitting e into two new edges (u, m) and (m, v) : $V(G_1^\bullet) = \{u, m, v\}$, $E(G_1^\bullet) = \{(u, m), (m, v)\}$. The discrete coboundaries in oriented form are $d_0 = \begin{pmatrix} -1 & 1 \end{pmatrix}$ on $X_0^0 = \mathbb{R}^2$, and $d_1 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$ on $X_1^0 = \mathbb{R}^3$. The bonding maps are $p_{1,0}^0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $p_{1,0}^1 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \end{pmatrix}$. The vertex Laplacians are

$$A_0^{\text{vertex}} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad A_1^{\text{vertex}} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

With the partition $B = \{u, v\}$, $I = \{m\}$, the block decomposition yields $A_{BB} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $A_{BI} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$, $A_{IB} = \begin{pmatrix} -1 & -1 \end{pmatrix}$, $A_{II} = (2)$. The Schur complement is

$$\text{Schur}(A_1^{\text{vertex}}) = A_{BB} - A_{BI}A_{II}^{-1}A_{IB} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} = \frac{1}{2}A_0^{\text{vertex}}.$$

This is exactly Theorem 2.1.

The defect of strict refinement-compatibility is

$$D_0 := p_{1,0}^0 A_1^{\text{vertex}} - \frac{1}{2} A_0^{\text{vertex}} p_{1,0}^0 = \begin{pmatrix} 1 & -1 & 0 \\ 0 & -1 & 1 \end{pmatrix} - \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -1 & \frac{1}{2} \\ \frac{1}{2} & -1 & \frac{1}{2} \end{pmatrix}.$$

Both rows of D_0 act on $\phi = (\phi(u), \phi(m), \phi(v))$ as $\frac{1}{2}\phi(u) - \phi(m) + \frac{1}{2}\phi(v) = \frac{1}{2}(\phi(u) + \phi(v)) - \phi(m)$, the “midpoint deficit” of ϕ at m . This vanishes precisely when $\phi(m) = \frac{1}{2}(\phi(u) + \phi(v))$, i.e. when ϕ is the harmonic extension of its boundary $(\phi(u), \phi(v))$, in agreement with Proposition 4.1(i). In particular, D_0 does *not* vanish on $\{\phi : \phi(m) = 0\}$ (which gives $D_0\phi = (\frac{1}{2}(\phi(u) + \phi(v)), \frac{1}{2}(\phi(u) + \phi(v)))$, generically non-zero), confirming Proposition 4.1(ii).

6.2 Tower II: triangle $\rightarrow$ barycentric edge-subdivision

Let $G_0^\bullet$ be the 3-cycle on $V(G_0^\bullet) = \{0, 1, 2\}$ with edges $e_{01} = (0, 1)$, $e_{12} = (1, 2)$, $e_{20} = (2, 0)$. Refine by inserting one midpoint per edge, m_{01}, m_{12}, m_{20} , giving $V(G_1^\bullet) = \{0, 1, 2, m_{01}, m_{12}, m_{20}\}$ ($|V(G_1^\bullet)| = 6$) and the six new edges $(0, m_{01}), (m_{01}, 1), (1, m_{12}), (m_{12}, 2), (2, m_{20}), (m_{20}, 0)$. With $B = \{0, 1, 2\}$, $I = \{m_{01}, m_{12}, m_{20}\}$ and the identification of bonding maps $p_{1,0}^0, p_{1,0}^1$ analogously to Tower I, direct computation (verified by `explore_defect.py`) yields

$$A_0^{\text{vertex}} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

and the Schur complement

$$\text{Schur}(A_1^{\text{vertex}}) = \begin{pmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & 1 \end{pmatrix} = \frac{1}{2}A_0^{\text{vertex}},$$

again confirming Theorem 2.1. Eigenvalues of the defect operator $\text{Schur}(A_1^{\text{vertex}}) - A_0^{\text{vertex}}$ on this tower are $\{0, -\frac{3}{2}, -\frac{3}{2}\}$, confirming the ordering $\text{Schur} \leq A_0^{\text{vertex}}$ (i.e. the unscaled Schur complement under-estimates the level-0 Laplacian; rescaling by 2^n at level n exactly compensates).

7 Cascade-mechanism update

The present paper updates the worked-invocation status table of [1] §3.5 for the imported P3/P4 finite-level refinement tower as follows. The consumer-API obligations themselves ([1] §3.4) are unchanged; the cascade-closure mechanism, Definition, Proposition, and Corollary of [1] are unchanged.

Table 1: Updated status of cascade-mechanism consumer-API obligations in the imported P3/P4 finite-level tower with the scaled curvature operator $\tilde{A}_n = 2^n A_n^{\text{vertex}}$.

Obligation	Status (this paper)	Where established
(O0) Standing analytic / admissibility hypotheses	Closed. Finite-dimensional Hilbert structure of X_n^0 , C^1 bonding maps, well-posed flow $e^{-t\tilde{A}_n}$.	[3]; recap in §1.
(O1) Top-rung fixed point and convergence	Closed at the trivial level (zero state always residual-zero); nontrivial selection reduces to: kernel of $\tilde{A}_N$ is the constants on connected components of $G_N^\bullet$, and convergence is automatic by spectral decomposition. The selection rule itself is out of scope per [1] §5.	Proposition 5.1.
(O2) Flow intertwining	Closed at the Schur / quadratic-form level only. Strict operator-level identity $p^0 \tilde{A}_{n+1} = \tilde{A}_n p^0$ remains false (Proposition 4.1); strict flow-level identity also remains false. The boundary-energy form satisfies $\text{Schur}(\tilde{A}_{n+1}) = \tilde{A}_n$ exactly (Theorem 2.1).	Theorem 2.1, Proposition 4.1.
(O3) Residual monotonicity	Closed. $\tilde{R}_n(p^0\psi) \leq \tilde{R}_{n+1}(\psi)$ for every $\psi \in X_{n+1}^0$, with equality iff ψ is the harmonic extension of its boundary.	Theorem 3.1, Corollary 3.2.

The discharge of (O3) and the partial discharge of (O2) are non-trivial relative to [1]’s original status report; the structural discovery is that the source’s coboundary factor- $\frac{1}{2}$ relation (2) lifts

to the curvature level via the Schur complement, with the scaled operator $\tilde{A}_n := 2^n A_n^{\text{vertex}}$ as the natural refinement-compatible object at the boundary-energy level.

8 Discussion

What this paper does. It proves that, with the scaled curvature operator $\tilde{A}_n = 2^n A_n^{\text{vertex}}$, the cascade-mechanism residual-monotonicity obligation (O3) holds for the imported P3/P4 finite-level refinement tower, and the flow-intertwining obligation (O2) holds at the boundary-energy / Schur level. The mechanism by which both happen is the same: the source’s coboundary factor- $\frac{1}{2}$ relation $p^1 d_{n+1} = \frac{1}{2} d_n p^0$ pulled forward through the boundary–interior decomposition.

What this paper does not do. It does not establish strict operator-level $p^0 \tilde{A}_{n+1} = \tilde{A}_n p^0$ intertwining; that identity is genuinely false on the cascade refinement tower with new-midpoint refinement. The defect operator D_n defined in Proposition 4.1 vanishes precisely on the harmonic-extension subspace and is generically non-zero elsewhere (including on the level- n -supported sector $\{\psi : \psi_I = 0\}$). In particular, the strict cascade-mechanism flow-intertwining hypothesis (O2) at the operator/flow level remains a stronger condition than what the P3/P4 tower supports; cascade-mechanism users invoking the boundary-energy version of (O2) will find it discharged here, but those requiring strict operator-level intertwining will find the defect computed but not zero. The paper does not derive a selection rule for nontrivial elements of $\ker \tilde{A}_N$, in keeping with [1] §5’s explicit boundary.

Implication for cascade-mechanism v2.1. The minimum cascade-mechanism update is a single sentence in [1] §3.5 replacing the “(O3) open” line with a citation of Corollary 3.2 above, and a short remark recording the partial discharge of (O2) at the Schur level. The cascade-mechanism consumer API itself ([1] §3.4 (O0)–(O3)) does not change. The reproduction bundle of [1] is unaffected.

Open follow-ups. By Proposition 4.1(i), the strict identity $p^0 \tilde{A}_{n+1} = \tilde{A}_n p^0$ holds exactly on the harmonic-extension subspace $\text{range}(H_{n+1})$. The natural follow-up is whether this is the only invariant subspace on which strict intertwining holds (the negative answer on $\{\phi : \phi_I = 0\}$ is recorded above; a σ -fixed sector of the cascade σ -attractor is a possible richer candidate). A separate question is whether the cascade-mechanism flow $e^{-t\tilde{A}_{n+1}}$ preserves $\text{range}(H_{n+1})$ (numerically, it does not, since $\tilde{A}_{n+1}$ does not preserve the harmonic subspace). A separate question is whether a non-linear cascade-flow modification (gradient flow on a modified residual that absorbs the defect into the dynamics) would preserve the mechanism’s structure while admitting strict intertwining; this is out of scope for the present paper.

9 Reproducibility

All computations in §§2–6 are verified to exact rational precision (Fraction arithmetic) by the script `papers/cascade-refinement-compat/scripts/explore_defect.py`. A reproducer with frozen reference outputs lives at `papers/cascade-refinement-compat/repro/`; from the repository root, one invocation

```
cd papers/cascade-refinement-compat/repro
bash run_audit.sh
```

re-runs the verification on Towers I and II and diffs against the frozen expected outputs. A clean run prints `REFINEMENT COMPAT AUDIT PASSED`.

References

- [1] Lee Smart. Cascade closure: A conditional compatibility template for geometric state selection. Preprint, Institute of Vibrational Field Dynamics. Local source: `papers/cascade-mechanism/cascade-mechanism.tex`. The consumer API (O0)–(O3) discharged in part by the present paper is in `cascade-mechanism.tex` §3.4., 2026.
- [2] Lee Smart. Cascade closure dynamics: Finite closure functionals, gradient flow, and refinement compatibility. Preprint, Institute of Vibrational Field Dynamics. Local source: `papers/cascade-closure-dynamics/cascade-closure-dynamics.tex`. The `coboundary-refinement` relation $p^1 d_{n+1} = \frac{1}{2} d_n p^0$ used as the load-bearing source identity is in this paper at lines 896–937 (Proposition)., 2026.
- [3] Lee Smart. Cascade refinement spaces: Finite-level state spaces and bonding maps. Preprint, Institute of Vibrational Field Dynamics. Local source: `papers/cascade-refinement-spaces/cascade-refinement-spaces.tex`., 2026.